



MANIPAL

ACADEMY *of* HIGHER EDUCATION

(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

Linux OS and Scripting – Class 2

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MANIPAL SCHOOL OF INFORMATION SCIENCES

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Part A — Ownership Investigation

1. Create a directory named `ownership_lab`

2. Inside it, create the required files `cd ownership_lab; touch incident_report.txt`

`network.log credentials.txt`

3. Display detailed information for these files

`ls -l`

4. For each file, identify: Owner, Group, Owner permissions, Group permissions, Others permissions

total 0

`-rw-rw-r-- 1 haris haris 0 Sep 4 15:21 credentials.txt`

`-rw-rw-r-- 1 haris haris 0 Sep 4 15:21 incident_report.txt`

`-rw-rw-r-- 1 haris haris 0 Sep 4 15:21 network.log`

5. Determine UID and GID of the current user

`uid=1000(haris) gid=1000(haris)`

`groups=1000(haris),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),111(lpadmin),114(lxd)`

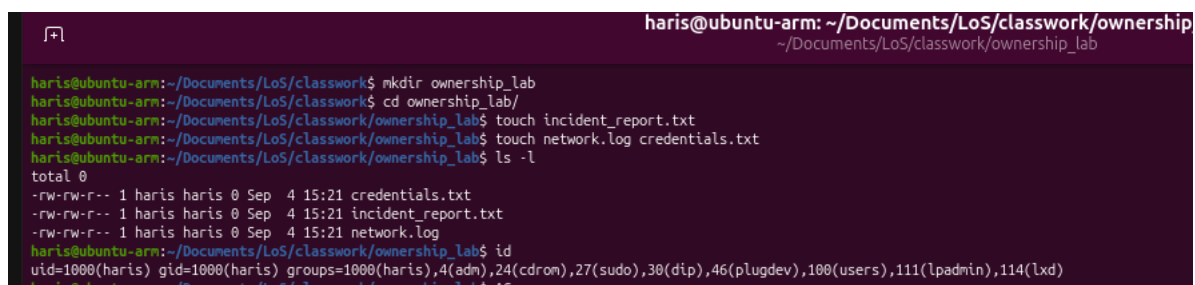
6. Compare the UID/GID information with the ownership displayed for the newly created files

When a file is created, it is normally assigned to the user who made it and to that user's primary group. The `'id'` command displays the user's UID and GID, while `'ls -l'` shows the file's owner and group.

7. Difference between file ownership and file permissions

File ownership tells us “who” the file belongs to by identifying its user and group.

File permissions tell you “what” those users can do, such as read, write, or execute the file.



```
haris@ubuntu-arm: ~/Documents/LoS/classwork/ownership
~/Documents/LoS/classwork/ownership_lab

haris@ubuntu-arm:~/Documents/LoS/classwork$ mkdir ownership_lab
haris@ubuntu-arm:~/Documents/LoS/classwork$ cd ownership_lab/
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ touch incident_report.txt
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ touch network.log credentials.txt
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ ls -l
total 0
-rw-rw-r-- 1 haris haris 0 Sep 4 15:21 credentials.txt
-rw-rw-r-- 1 haris haris 0 Sep 4 15:21 incident_report.txt
-rw-rw-r-- 1 haris haris 0 Sep 4 15:21 network.log
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ id
uid=1000(haris) gid=1000(haris) groups=1000(haris),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),111(lpadmin),114(lxd)
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$
```

Part B — chown: Changing Ownership

1. Change the owner of incident_report.txt to alice

sudo chown alice incident_report.txt

2. Change the owner of network.log to bob.

sudo chown bob network.log

3. Change the owner of credentials.txt to charlie.

sudo chown charlie credentials.txt

4. Verify each change.

ls -l

5. Try changing the owner of a file to another user without using sudo. Record the

result chown bob incident_report.txt

6. Explain why an ordinary user is generally not permitted to transfer file ownership to another user.

A regular user typically can't change a file's ownership because that could lead to security issues. For instance, a user might try to shift a file to another account to bypass storage limits or blame someone else for a malicious file. That's why ownership changes are usually reserved for administrators.

7. Using a single command, configure:

incident_report.txt Owner : alice Group :

security sudo chown alice:security

incident_report.txt

```
haris@ubuntu-arm: ~/Documents/LoS/classwork/ownership_lab
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ sudo chown alice incident_report.txt
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ sudo chown bob network.log
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ sudo chown charlie credentials.txt
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ ls -l
total 0
-rw-rw-r-- 1 charlie haris 0 Sep  4 15:21 credentials.txt
-rw-rw-r-- 1 alice   haris 0 Sep  4 15:21 incident_report.txt
-rw-rw-r-- 1 bob     haris 0 Sep  4 15:21 network.log
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ result chown bob incident_report.txt
result: command not found
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ chown bob incident_report.txt
chown: changing ownership of 'incident_report.txt': Operation not permitted (os error 1)
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ sudo chown bob incident_report.txt
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ ls -l
total 0
-rw-rw-r-- 1 charlie haris 0 Sep  4 15:21 credentials.txt
-rw-rw-r-- 1 bob     haris 0 Sep  4 15:21 incident_report.txt
-rw-rw-r-- 1 bob     haris 0 Sep  4 15:21 network.log
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ sudo chown alice incident_report.txt
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ ls -l
total 0
-rw-rw-r-- 1 charlie haris 0 Sep  4 15:21 credentials.txt
-rw-rw-r-- 1 alice   haris 0 Sep  4 15:21 incident_report.txt
-rw-rw-r-- 1 bob     haris 0 Sep  4 15:21 network.log
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$
```

```

-rw-rw-r-- 1 bob harris 0 Sep  4 15:21 network.log
harris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ sudo chown alice:security incident_report.txt
harris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ ls -l incident_report.txt
-rw-rw-r-- 1 alice security 0 Sep  4 15:21 incident_report.txt
harris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$

```

Part C — chgrp: Changing Group Ownership

1. Change the group ownership of network.log to developers.

```
sudo chgrp developers network.log
```

2. Change the group ownership of credentials.txt to security.

```
sudo chgrp security credentials.txt
```

3. Verify the changes.

```
ls -l
```

4. Display the group memberships of alice, bob, and charlie.

```
groups alice
```

```
groups bob
```

```
groups charlie
```

5. Log in as alice and create alice_report.txt.

```
sudo su - alice
```

```
cd ownership_lab
```

```
touch
```

```
alice_report.txt
```

6. Attempt to change the group of alice_report.txt to security.

```
chgrp security alice_report.txt
```

7. Attempt to change its group to developers.

```
chgrp developers alice_report.txt
```

8. Compare the results.

```
chgrp: changing group of 'alice_report.txt': Operation not permitted (os error 1)
```

9. Explain how a user's group membership affects their ability to change the group ownership of a file they own.

A regular user can change the group of a file they own only to a group they are already a member of. So, their group membership decides which groups they can assign without using admin privileges.

```

haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ sudo chgrp developers network.log
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ sudo chgrp security credentials.txt
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ ls -l
total 0
-rw-rw-r-- 1 charlie security  0 Sep  4 15:21 credentials.txt
-rw-rw-r-- 1 alice  security  0 Sep  4 15:21 incident_report.txt
-rw-rw-r-- 1 bob   developers 0 Sep  4 15:21 network.log
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ groups alice
alice : alice users security
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ groups bob
bob : bob users developers security
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ groups charlie
charlie : charlie users security
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ sudo su - alice
alice@ubuntu-arm:~$ cd ownership_lab
-bash: cd: ownership_lab: No such file or directory
alice@ubuntu-arm:~$ touch alice_report.txt
alice@ubuntu-arm:~$ chgrp security alice_report.txt
alice@ubuntu-arm:~$ chgrp developers alice_report.txt
chgrp: changing group of 'alice_report.txt': Operation not permitted (os error 1)
alice@ubuntu-arm:~$

```

Part D — Ownership + Permission Investigation

1. Configure -rw-r----- alice security incident_report.txt sudo chmod 640

incident_report.txt

2. Identify what operations are available to: alice, charlie, bob sudo usermod -aG

security charlie

3. A user who belongs to neither security nor the file owner charlie belongs to security.

Record the result of each operation. Test whether Charlie can:

- Read incident_report.txt
- Modify incident_report.txt
- Delete incident_report.txt

sudo chmod 660

incident_report.txt

sudo su - charlie

cd ownership_lab

```

haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ sudo chmod 640 incident_report.txt
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ sudo usermod -aG security charlie
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ sudo chmod 660 incident_report.txt
haris@ubuntu-arm:~/Documents/LoS/classwork/ownership_lab$ sudo su - charlie
charlie@ubuntu-arm:~$ ls
charlie@ubuntu-arm:~$ cd ownership_lab
-bash: cd: ownership_lab: No such file or directory
charlie@ubuntu-arm:~$ ls -l
total 0
charlie@ubuntu-arm:~$ cd /home/haris/ownership_lab
-bash: cd: /home/haris/ownership_lab: Permission denied
charlie@ubuntu-arm:~$ cat incident_report.txt

```

echo "test" >> incident_report.txt

rm incident_report.txt -> can't

4. Explain why Charlie's access changed even though the owner and group of the file did not change. Remove bob from the security group.

charlie's access changed because the group permissions were updated from read-only (r--) to read and write (rw-). Since the group stayed security, Charlie, as a member of that group, automatically gained write access.

5. Explain why Bob's effective permissions changed.

Linux checks permissions in this order: *user, then group, then others*. While Bob was a member of the 'security' group, he received the group permissions ('rw-'). After he was removed from that group, group access no longer applied, so Linux fell back to the others permissions ('---'), giving him no access.